CS621/CSL622
Quantum Computing For Computer Scientists

Quantum Architecture

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Reversible Gates

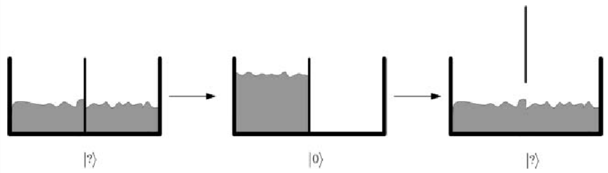
Reversibility of Writing

- Reversible gates predates the idea of quantum computing
  'In the 1960s, Rolf Landauer analyzed computational processes and showed that erasing information, as opposed to writing information, is what causes energy loss and heat.'

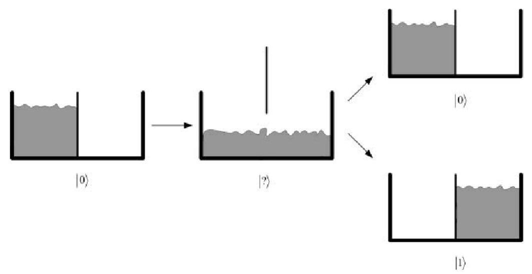
  Landauer's principle

- Losing information means energy is being dissipated

- Writing is reversible.

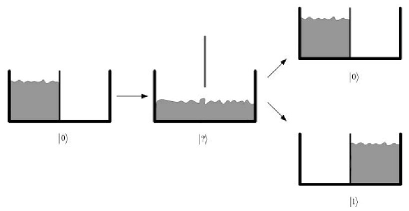


- **Erasing** information is an irreversible, energy-dissipating operation. How?



- How could we return to the original state?
- There are two possible states to return to.

- The original state was $|0\rangle$
- Why can't we go back to it?

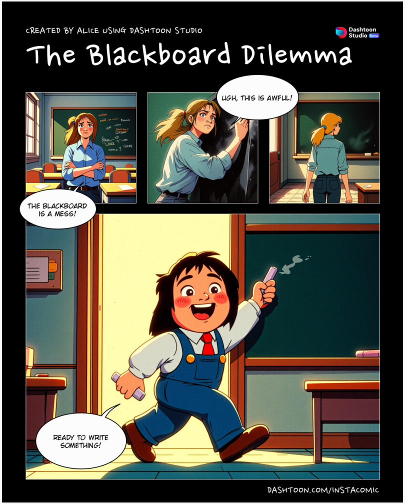


Implies that

- Information is copied to the brain.
- The system is both the tub and the brain
- Erasing of $|0\rangle$ was not complete
- Our brain was still storing the information.

- If Alice writes a letter on an empty blackboard and then Bob walks into the room, he can then erase the letter that Alice wrote on the board and return the blackboard into its original pristine state.
- Thus, writing is reversible.
- In contrast, if there is a board with writing on it and Alice erases the board, then when Bob walks into the room he cannot write what was on the board. Bob does not know what was on the board before Alice erased it.
- So Alice's erasing was not reversible.


CREATED BY ALICE USING DASHTOON STUDIO
Dashtoon Studio Beta
The Disappearing Message
LET ME WRITE SOMETHING ON THIS BOARD.
HELLO
PERFECT! A MESSAGE FOR BOB.
HELLO
LET'S GET THIS BOARD CLEAN.
WHERE DID MY MESSAGE GO?
DASHTOON.COM/INSTACOMIC


CREATED BY ALICE USING DASHTOON STUDIO
Dashtoon Studio Beta
The Blackboard Dilemma
THE BLACKBOARD IS A MESS!
UGH, THIS IS AWFUL!
READY TO WRITE SOMETHING!
DASHTOON.COM/INSTACOMIC

Point-to-Ponder

If erasing information is the only operation that uses energy, then a computer that is reversible and does not erase, would not use any energy.

- **Charles H. Bennett** was first known to have started working on reversible circuits and programs in the 1970s.

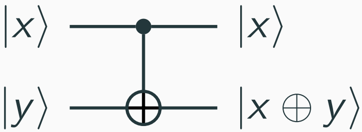
Are Classical Gates Reversible?

- The NOT gate and the identity gates are reversible.
- They are their own inverses

$$NOT \times NOT = I_2 \qquad I_n \times I_n = I_n$$

- What about AND?
- Given an output of $|0\rangle$ from AND, one cannot determine if the input was $|00\rangle$, $|01\rangle$, or $|10\rangle$

Controlled-Not Gate

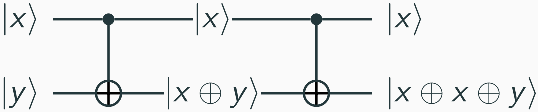


$$\begin{array}{c} \\ \mathbf{00} \\ \mathbf{01} \\ \mathbf{10} \\ \mathbf{11} \end{array}\begin{array}{c} \begin{array}{cccc} \mathbf{00} & \mathbf{01} & \mathbf{10} & \mathbf{11} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}.$$

- Gate has two inputs and two output
- The top input is the **control** bit
- The bottom input is the **target** bit
- cNOT gate: $|x, y\rangle \to |x, x \oplus y\rangle$
- Now compute cNot $|10\rangle$

cNOT is Reversible

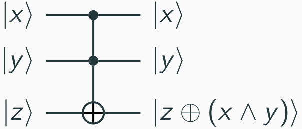
- cNOT $\times$ cNOT



- Verify the same using the cNOT matrix.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} .$$

ccNOT — Toffoli Gate



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\begin{bmatrix}
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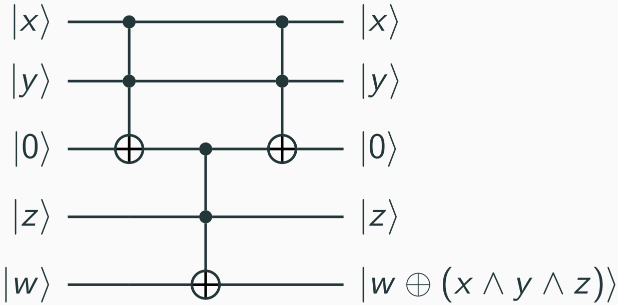
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{bmatrix}
\end{array}
$$

- Similar to the controlled-NOT gate
- But with **two** controlling bits
- Target bit flips only when both controlling bits are in state $|1\rangle$
- Toffoli gate: $|x, y, z\rangle \rightarrow |x, y, z \oplus (x \wedge y)\rangle$
- The Toffoli gate is also reversible

A Derived Gate

- A gate with three controlling bits can be constructed from three Toffoli gates



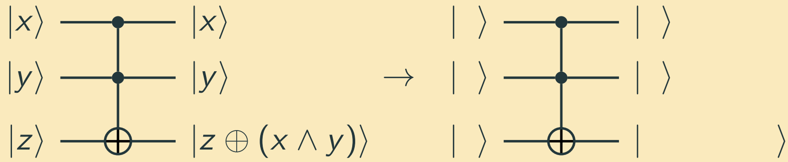
Toffoli - Universal Gate

- Any logical gate can be derived using copies of the Toffoli gates
- This makes the Toffoli gate an universal gate
- One can make a reversible computer using only Toffoli gates
- Such a computer would, in theory, neither use any energy nor give off any heat.

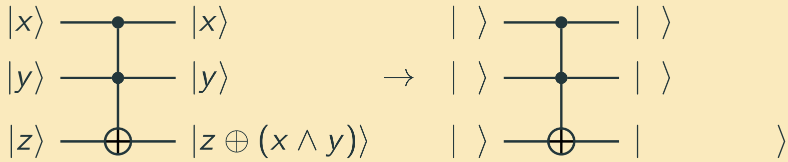
How to show that the Toffoli gate is universal?

One needs to show that the Toffoli gate can be used to make both the AND and NOT gates.

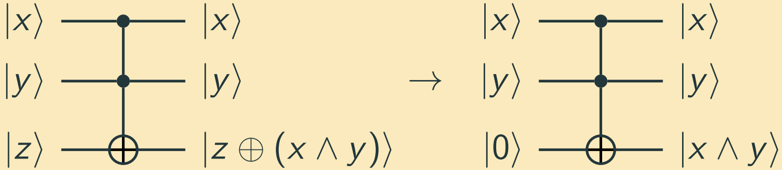
Toffoli $\rightarrow$ AND



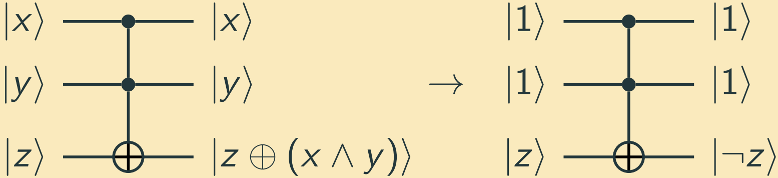
Toffoli $\rightarrow$ NOT



Toffoli $\to$ AND



Toffoli $\to$ NOT



Trick to Construct All Gates

- A way of producing a fanout of values
- A gate is needed that inputs a value and outputs two of the same values

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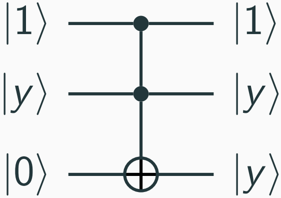
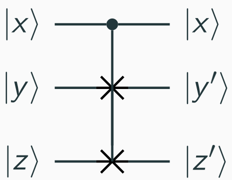


Exercise

1. Construct the NAND with one Toffoli gate.
2. Construct the XOR with one Toffoli gate.
3. Construct the OR gate with two Toffoli gates.

Conditional Swap



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\mathbf{000}\\ \mathbf{001}\\ \mathbf{010}\\ \mathbf{011}\\ \mathbf{100}\\ \mathbf{101}\\ \mathbf{110}\\ \mathbf{111}
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\end{array}\\
\begin{bmatrix}
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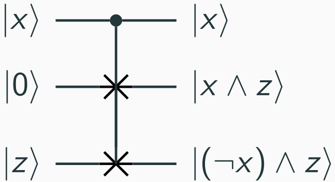
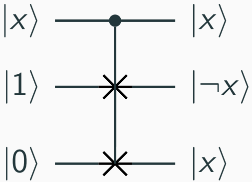
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
$$

- Top $|x\rangle$ input is control input
- If $|x\rangle$ is set to $|0\rangle$, then $|y'\rangle = |y\rangle$ and $|z'\rangle = |z\rangle$
- If $|x\rangle$ is set to $|1\rangle$, then $|y'\rangle = |z\rangle$ and $|z'\rangle = |y\rangle$
- Fredkin gate: $|0, y, z\rangle \rightarrow |0, y, z\rangle$ and $|1, y, z\rangle \rightarrow |1, z, y\rangle$
- The Fredkin gate is also reversible

Fredkin Gate is also Universal



Fredkin $\rightarrow$ AND



Fredkin $\rightarrow$ NOT

Point-to-Ponder

$$\begin{array}{cc}
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0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
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\end{bmatrix}
\end{array}$$

- Toffoli and Fredkin gates are not only reversible gates; a glance at their matrices indicates that they are also **unitary**
- Recall the significance of unitary matrices with regards to operations in quantum theory
- Points us in the direction of their usage as quantum gates

Quantum Gates

Definition (Quantum Gate)

A *quantum gate* is simply an operator that acts on qubits. Such operators will be represented by unitary matrices.

- Identity operator I
- Hadamard gate H
- NOT gate
- Controlled-NOT gate
- Toffoli gate
- Fredkin gate

- Some other[1] quantum gates: **Pauli** matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Verify if each of the quantum gates have unitary matrices

[1] There are other quantum gates too which we will encounter later.

Experimentally

A qubit's state can be changed through some **physical** action such as applying an **electromagnetic laser** or passing it through an **optical device**.

Mathematically

This corresponds to multiplying the qubit $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ by some unitary matrix U so that after the operation the state is now $|\psi'\rangle = U |\psi\rangle$.

- The physical action of interacting with the state corresponds mathematically to applying a unitary operator.

- **Unitary** is a mathematical term which expresses that U can only act on the qubit in such a way that the **total probability** $|\alpha|^2 + |\beta|^2$ does not change.
- In all mathematical constructions of quantum mechanics, one fundamental assumption is that each (matrix) operator must be **unitary**.

Significance of Unitary Matrix Assumption

- Ensures that after changing any state through an action, the total probability to observe all possible states will still add up to 100%
- If this did not happen, then we could not interpret the results of quantum mechanics to be probabilistic[2].

[2] The results would disagree with the many experiments that have been performed to date

- A matrix is unitary if and only if it preserves the **Euclidean norm**.
- There is a very simple condition to check this:

Definition (Unitary)

A matrix U is unitary if and only if

$$U^\dagger U = UU^\dagger = I,$$

where $U^\dagger$ is the **conjugate transpose** of U

To take the **conjugate transpose** of U

- take the transpose of U
- then take the complex conjugate of each of the entries

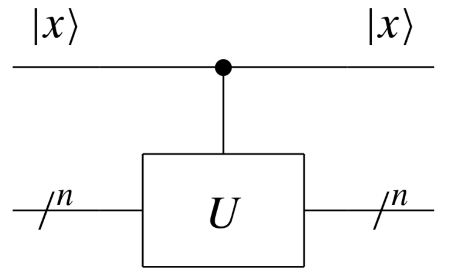
Exercise

- What is the conjugate transpose of the following matrix?

$$A = \begin{pmatrix} 1 & i \\ 1 & i \end{pmatrix} \qquad A^\dagger = \begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}$$

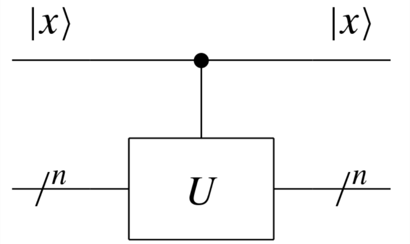
- Is it unitary?

- If a certain (qu)bit is true, then a particular operation should be performed, otherwise the operation is not performed.
- For every $n-$qubit unitary operation U, we can create a unitary $(n+1)-$qubit operation controlled$-U$ or ^{C}U.



- This operation will perform the U operation if the top $|x\rangle$ input is a $|1\rangle$ and will simply perform the identity operation if $|x\rangle$ is $|0\rangle$.

controlled$-U$ Matrix



If, $\quad U = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad$ Then, $\quad {}^{C}U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}$

- This is not unitary or, in general, even reversible.
- This operation is usually performed at the end of a computation when we want to measure qubits (and find bits).
- Following symbols are generally used to denote a measurement.

- We will learn about measurements and partial measurement in the upcoming parts of the course.

References

1. Quantum Computing for Computer Scientists, by Noson S. Yanofsky, Mirco A. Mannucci
2. Quantum Computing Explained, David Mcmahon. John Wiley & Sons
3. Lecture Notes on Quantum Computation, John Watrous, University of Calgary
    - `https://cs.uwaterloo.ca/~watrous/QC-notes/`